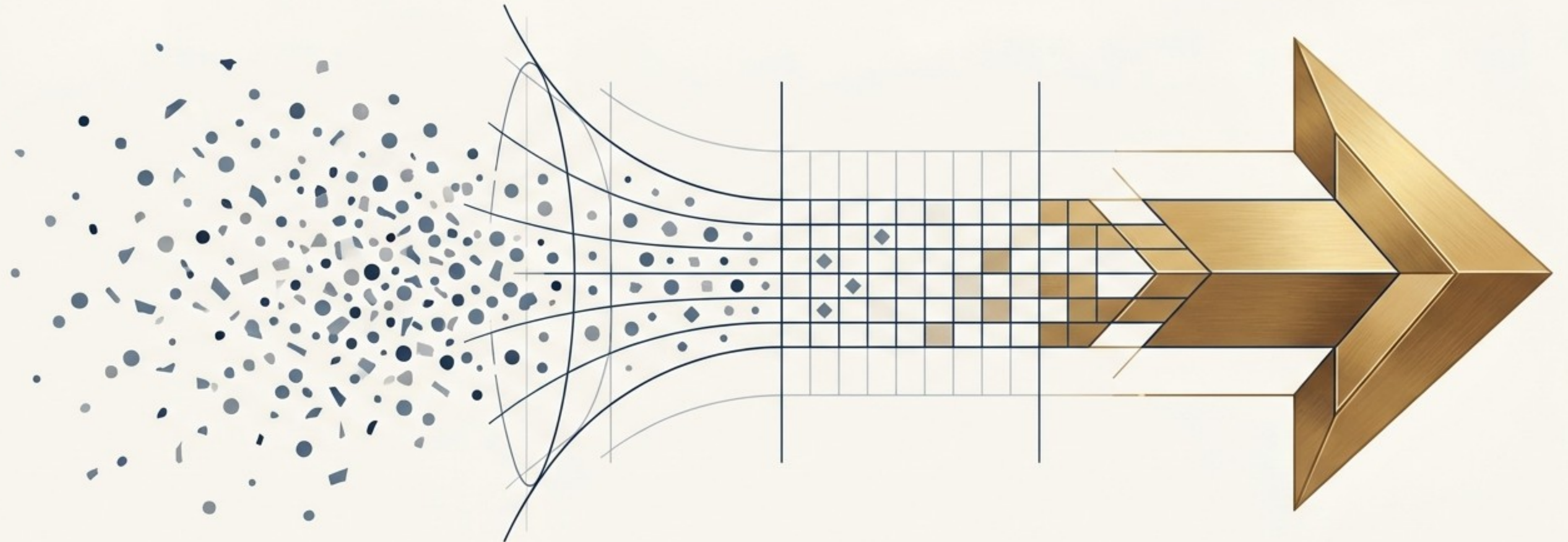


BikeStore: From Raw Data to Business Analytics

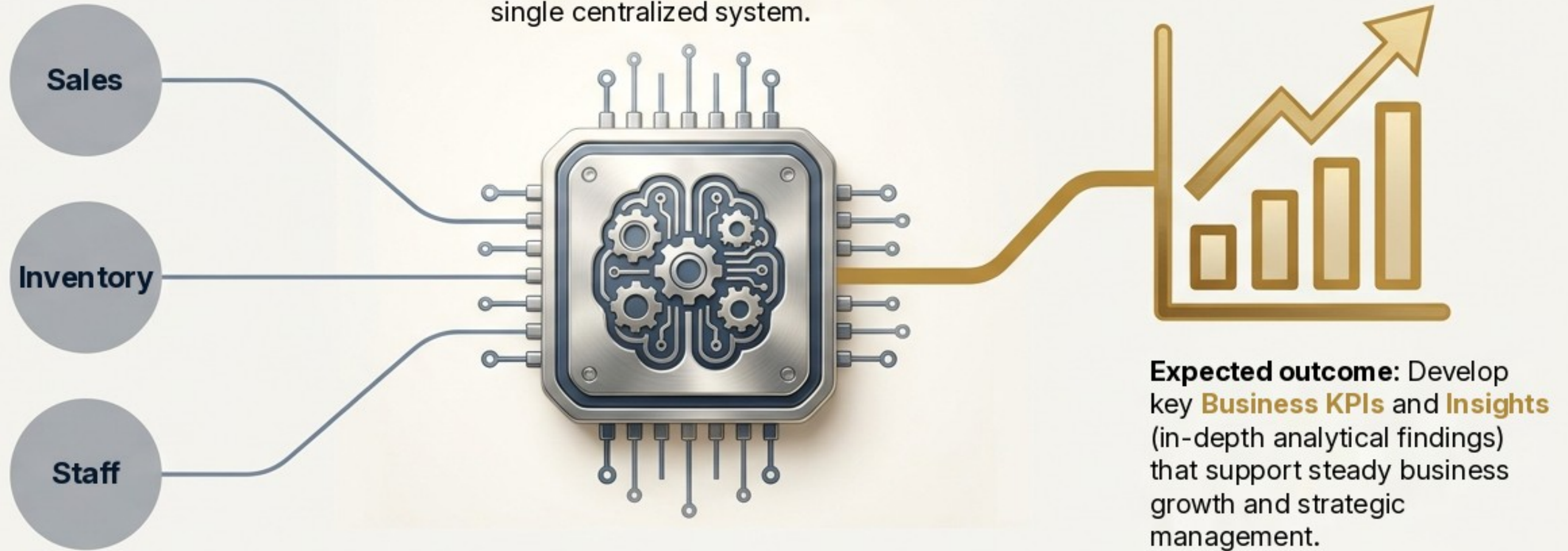
Raw Data → SQL → Insight → Decision



Project Objective: From Silos to a Single System

Subject: BikeStore — a retail business data analysis project.

Main goal: Consolidate all of the company's fragmented data into a single centralized system.



Data Exists, but Is Hard to Analyze

Key Thesis: Because the data was stored as raw transactions, it was difficult to see the main indicators quickly.



From Chaos to Order: Architecture

How did the data start working together? All departments (Sales, Stores, Products, Inventory, Staff, Customers, Regions) were linked to one another and organized into a single system.



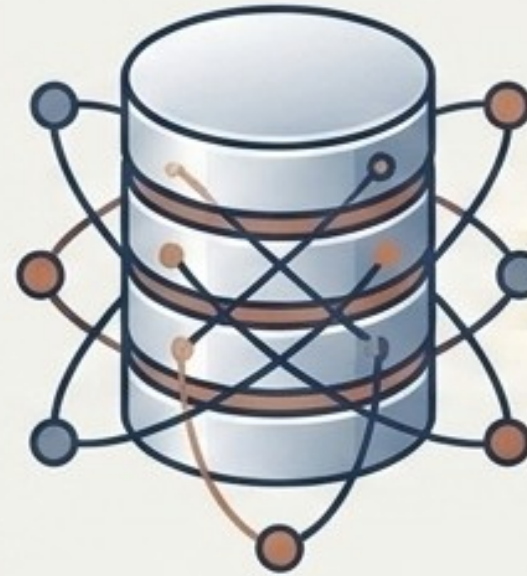
1. 9 CSV Files

Scattered, unrelated raw tables.



2. SQL Server

Loading the data into a centralized database.



3. Structured Database

Building a network of relational tables.



4. Analysis

Management-ready analytical system.

Two Ways to Analyze Data

Key Thesis: Two core SQL tools were implemented for data analysis. Together they cover the business's different needs.



7 Views (Continuous Analysis)

- 1. Purpose:** Provide frequently used KPIs in a ready-to-use form.
- 2. Mechanism:** When a view is called, it returns results based on the current data in the database.
- 3. Business Example:** Daily revenue and overall stock status.



5 Stored Procedures (Parameterized Analysis)

- 1. Purpose:** Get precise analysis for a specific parameter.
- 2. Mechanism:** Get results by passing a parameter such as store, customer, product or year.
- 3. Business Example:** Sales analysis of a single category for one store in the previous year.

Key Indicators Obtained with SQL

Key Thesis: Key KPIs for sales, orders, inventory and products were calculated using SQL.

Revenue

Total revenue and sales trends.



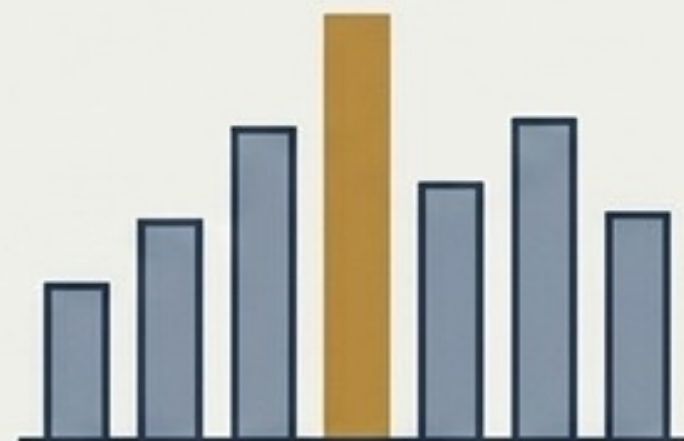
Orders & AOV

Number of orders and Average Order Value (AOV).



Sales Volume

Change in sales volume over time.



Stock Status

Low Stock products and inventory status.



Key Findings Identified in the Analysis

Key Thesis: SQL analyses revealed differences across inventory, stores, products, staff and regions.

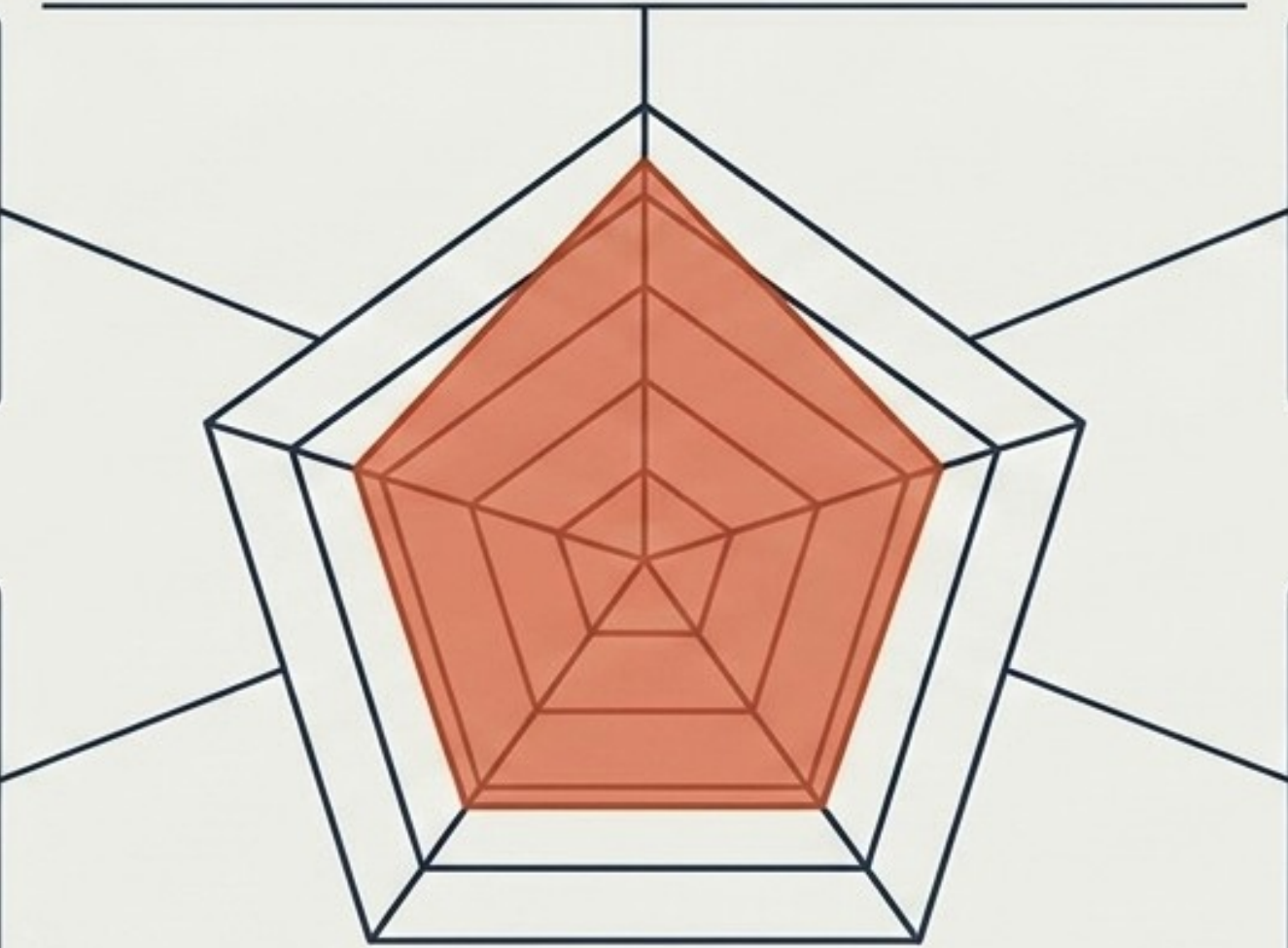
1. Low Stock: Products whose stock is below the defined threshold.

2. Weak Stores:
Stores with low results in Revenue or Orders.

3. Weak Products:
Products with low results in Sales Volume or Revenue.

5. Regional Differences:
Differences in sales results across regions.

4. Staff Performance:
Staff results differ in Orders and Revenue.



Insight to Action: What Should We Do Next?

Key Thesis: Seeing the problem is not enough; it has to be solved. Analyses translate directly into Business Actions:

Low Stock
(Low inventory)



Restock on time to prevent lost sales.

Weak Store
(Underperformer)



Investigate in depth (Performance Analysis) and reallocate resources.

Top Products
(Best sellers)



Guarantee constant availability in sales and prioritize them in marketing.

Top Staff
(Star performers)



Standardize their working approach (Best Practices) and train others.

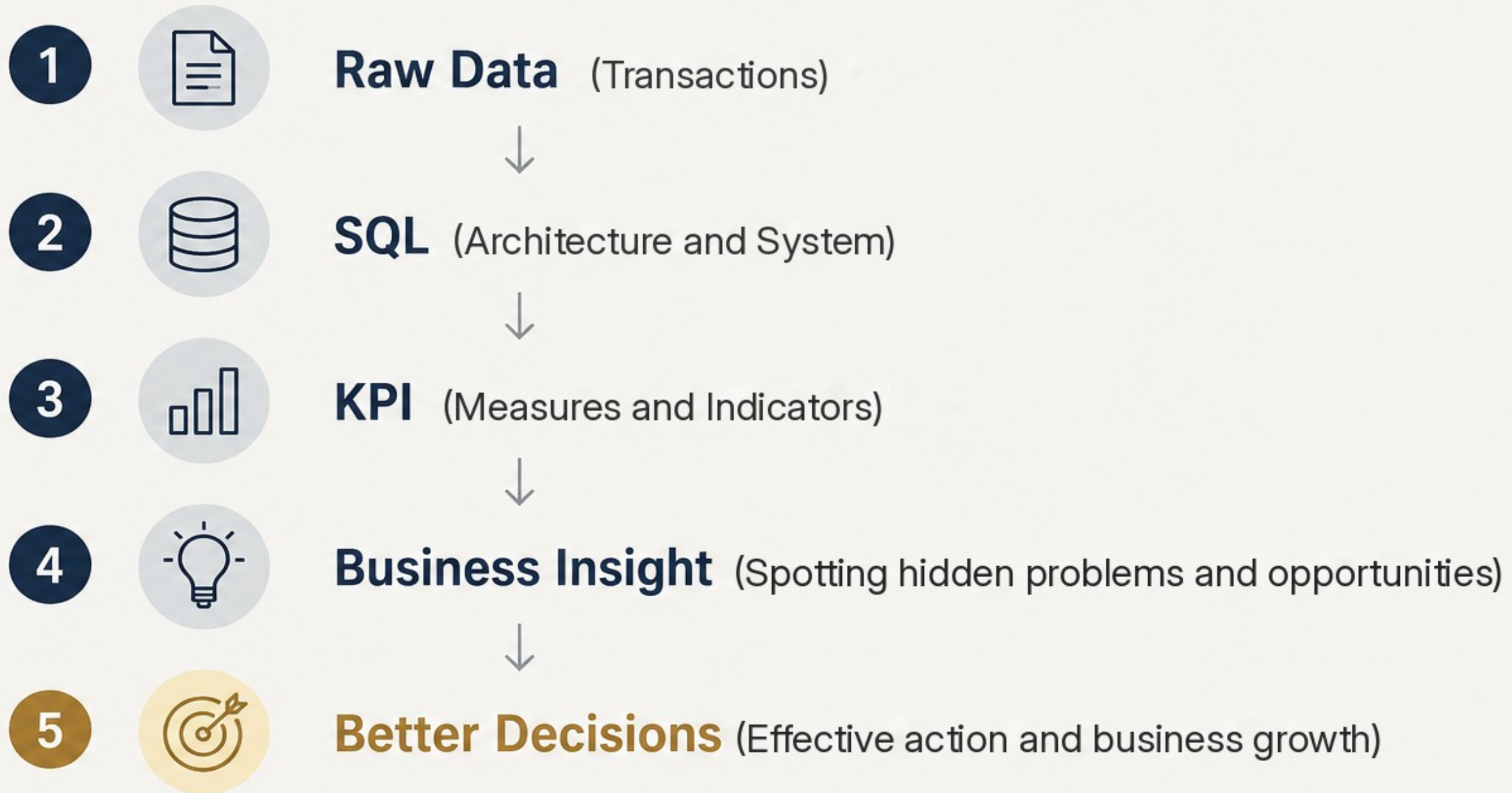
Next Steps: Evolution

Key Thesis: The project does not end with SQL. The plan is to take the system to a smarter, more automated level on top of the foundation built:



Conclusion: The Path from Data to Decision

The final formula of the entire process:



THANK YOU!

We appreciate your attention

...

Yunusov Javohir

Group F67



BikeStore



SQL Server